

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

Explain the difference between distance and displacement using a closed-loop journey.

[4 marks]

2

Calculate the current through a $12\ \Omega$ resistor connected across a 9 V supply, and the power dissipated.

[5 marks]

$$V = IR$$

3

Describe how refraction changes wavelength but not frequency, and explain why.

[5 marks]

4

A wave has frequency 250 Hz and wavelength 1.4 m. Calculate its speed and state the equation used.

[4 marks]

$$v = f\lambda$$

5

Explain why sound cannot be polarised but light can, referring to the oscillation direction in each case.

[5 marks]

6

A trolley accelerates uniformly from rest. Derive the distance travelled in time t from first principles.

[6 marks]

$$s = ut + \frac{1}{2}at^2$$

7

State the principle of conservation of momentum and apply it to a head-on elastic collision.

[6 marks]

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

8

A resistor network has two $6\ \Omega$ resistors in parallel with a $3\ \Omega$ resistor in series. Find the total resistance.

[5 marks]